

Faculty Research Grant & Publication Tracker

1. Introduction & Scenario Review

The research deanery requires an automated portal to track sponsored research grants, faculty publication metrics (indexing, citations), co-author approval workflows, and fund burn-up analytics.

2. Elicit & Draft Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-1	Functional	The system shall track fund allocations and show fund burn-up analytics	High	Each ongoing research process shows current fund, expenses and burn analytics	Dean should be able to see research statistics
FR-2	Functional	Faculty dean can grant fund allocations requested by research faculty	High	Fund allocation does not go through until approval.	Necessary wall before funds are allocated
FR-3	Functional	Faculty can log equipment procurement expenses, and overall budget	High	Balance updates accurately	Expense tracking
FR-4	Functional	Faculty can request co-author credit, to be approved by author-and-dean.	Medium	A faculty can request co-author credit on any research paper, it does not go through until author of the paper approves	Co-author functionality
FR-5	Functional	Faculty can log publication metrics (indexing, citations)	Medium	Each research project will have indexes and citations that authors can update	Publication detail tracking
NFR-1	Non-Functional	The system shall maintain an immutable, tamper-evident audit ledger for all financial	High	Once logged, fund allocations and expenditure should not be alterable by anyone	Financial records should be accurate and immutable

		approvals and journal status modifications.			
NFR-2	Non-Functional	Expenses should not exceed the granted funds.	High	Expenses do not log if they are exceeding funds	Non-existent funds shouldn't be used up.

3. Peer Critique & Revision:

The dean need not approve co-author credit, that can be resolved between authors.

4. Extract Actors & Use Cases:

Actors:

1. **Faculty Researcher:** works on projects. Needs to:
 - Request fund allocations
 - Log equipment expenses
2. **Author:** Publishes papers. Needs to:
 - Log publication metrics
 - Request/approve co-author credit
3. **Research Dean:** Tracks expenses and allocates funds. Needs to:
 - Track expenses and see burn-up rate for each project
 - Approve/reject fund allocations

Use-cases:

1. Request Research Grant Allocation

- **Primary Actor:** Faculty Researcher
- **Secondary Actor:** System

2. Review and Approve/Reject Grant Allocations

- **Primary Actor:** Research Dean
- **Secondary Actor:** Faculty Researcher, System

3. Log Equipment Procurement & Expenses

- **Primary Actor:** Faculty Researcher
- **Secondary Actor:** System

4. Monitor Fund Burn-Up Analytics & Financial Status

- **Primary Actor:** Research Dean
- **Secondary Actor:** System

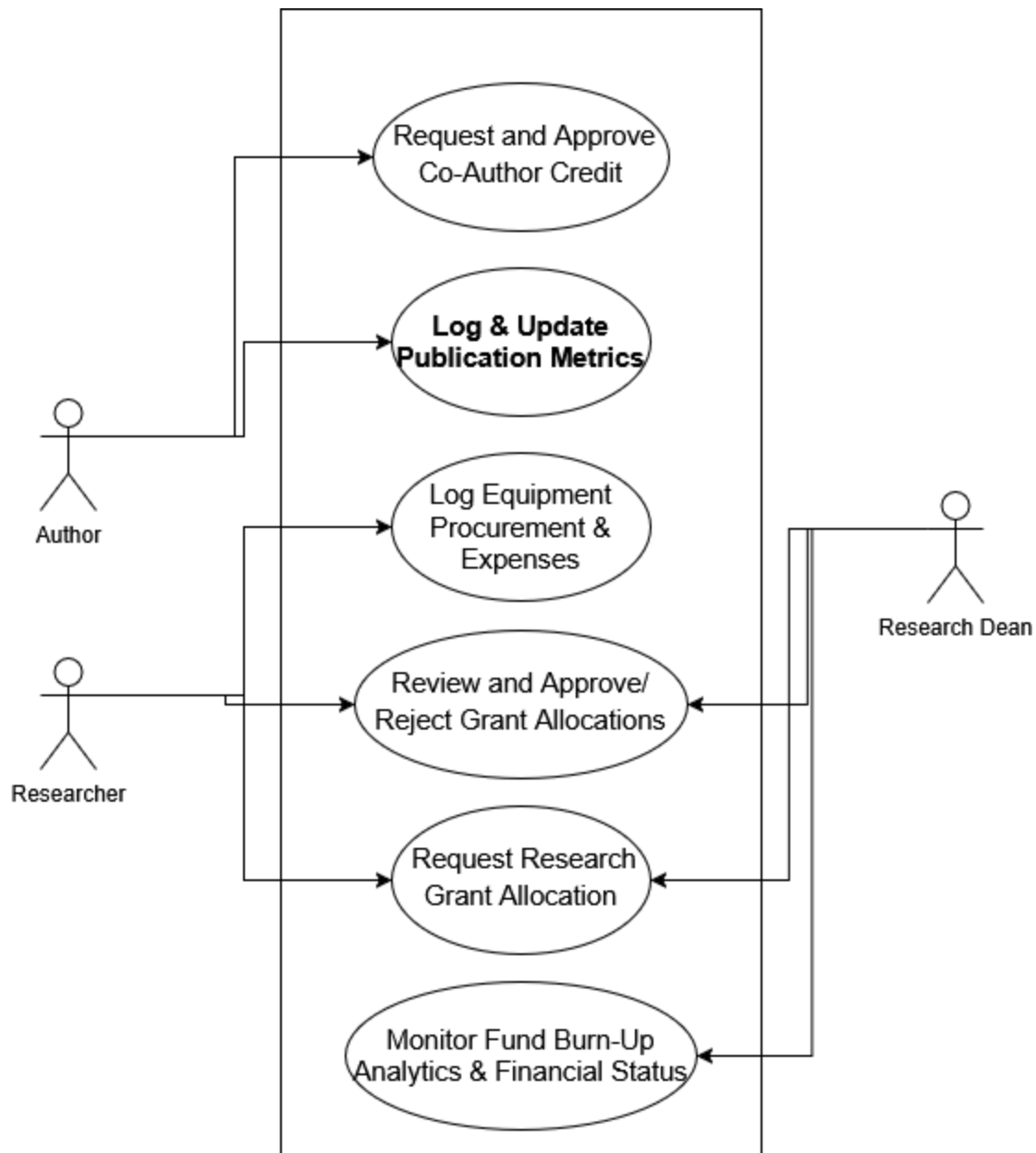
5. Log & Update Publication Metrics

- **Primary Actor:** Author (Faculty)
- **Secondary Actor:** System

6. Request and Approve Co-Author Credit

- **Primary Actor:** Faculty Researcher (Requesting Co-Author)
- **Secondary Actor:** Primary Author (Approving Faculty)
- **Precondition:** A paper has been logged in the system by the Primary Author.

5. Draw at least one UML Use-Case Diagram:



6. Write One Use-Case Flow:

2. Review and Approve/Reject Grant Allocations

- **Primary Actor:** Research Dean
- **Secondary Actor:** Faculty Researcher, System
- **Precondition:** Pending fund allocation requests exist in the system.
- **Trigger:** The Dean accesses the pending approvals dashboard.
- **Main Success Scenario:**
 1. Dean reviews the grant request details, budget breakdown, and available deanery funds.

2. Dean approves the allocation.
 3. System credits the approved amount to the project's total balance.
 4. System logs the immutable financial approval entry in the audit ledger.
 5. System notifies the Faculty Researcher of the approval.
- **Alternative Scenario (Rejection):**
 1. Dean rejects the request with comments.
 2. System updates the status to "Rejected" and notifies the researcher without updating project funds.
 - **Postcondition:** Project fund balance is updated, and the transaction is recorded permanently.